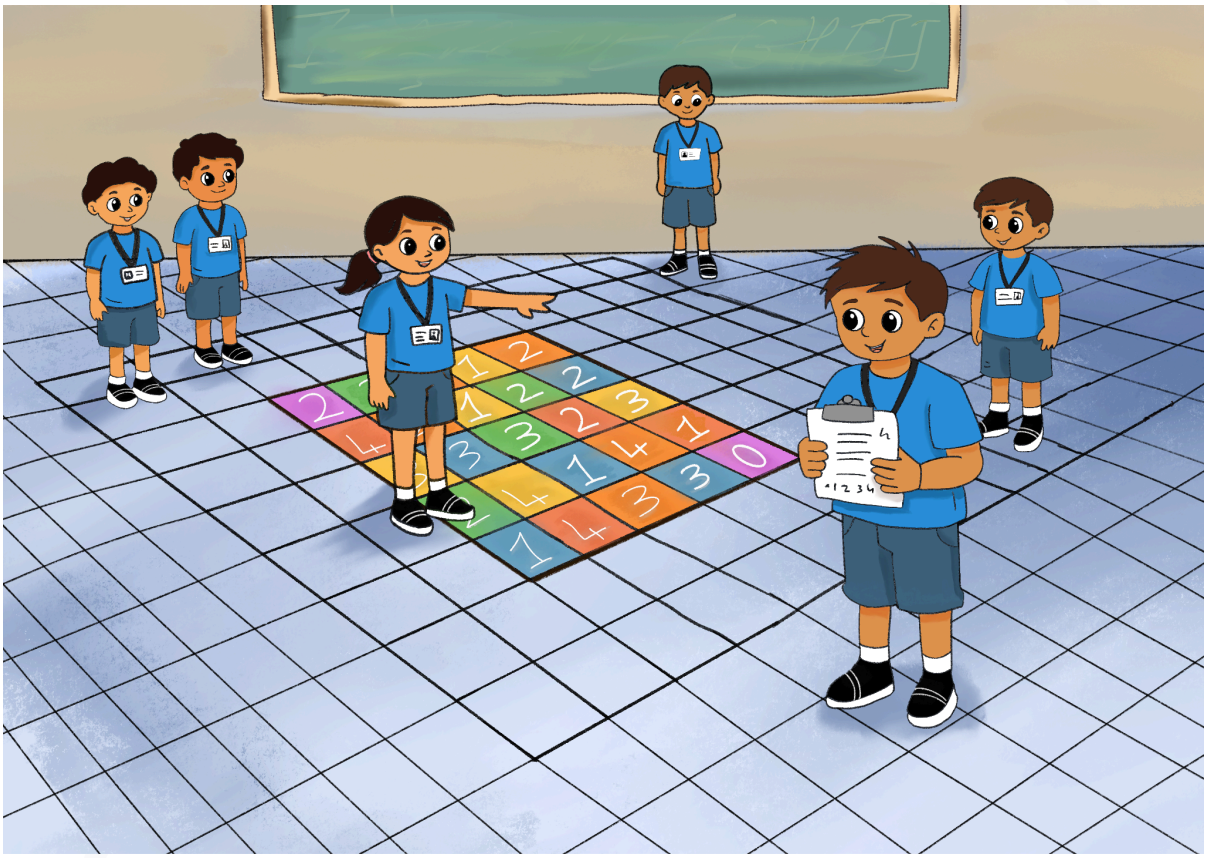
Number Maze

Grade 5: Maze Solving

Introduction

In these activities, students explore what a computer program is by writing a set of steps to solve a maze. A program is basically a set of instructions to tell the computer what to do. To ensure the program is working and solving the problem, it must also be tested. The activity provides students with an opportunity to play different roles and understand the basics of computer programming.



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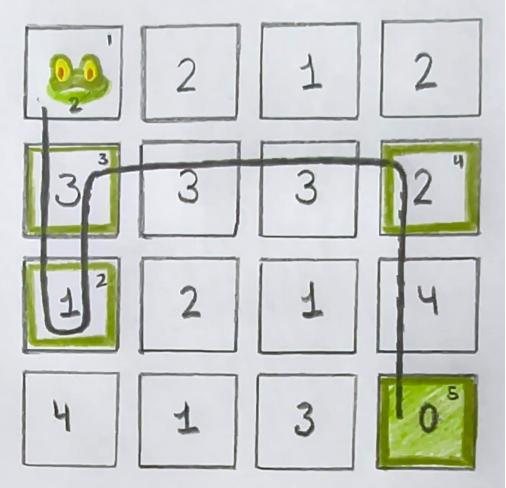
Activity	Time	Description
Launch	10-15 mins	The teacher will take the students outside and explain the number-maze puzzle. First, the students will solve the puzzle. Next, they have to write step-by-step instructions for one of their group members. Templates of the Number-grid maze puzzle are at the end.
Trails by Students	10-15 min	Student's tryout to solve the number-maze puzzles in groups of 4 or 5 and create instructions to reach from start to end. One of the students will follow the instructions in the maze to see if it's working correctly.
Discussions and Explorations	10 min	Discussion on the activity about how it's related to computer programming: Solving, converting to a set of instructions, and testing the instructions.

Computational Thinking Components

- *Algorithmic Thinking*. Students are learning to sequence a set of instructions to complete the puzzle.
- *Abstraction*. The 'programming' students will be doing in these activities is an abstract version of the computer programming language.

Activity: Number-Maze solving and Programming

The puzzle is to start from the top-left square position and reach the bottom-right square. Each square has the number written on it. Students can jump in either direction (top, down, right, or left) equal to the number on the square. Consider the example shown below.



The 4x4 grid has numbers written in each square. The starting position (top-left square) has the number 2 written on it. So, you need to jump down two squares. You will reach the square with the number 1 on it (3rd row, 1st column). Then you jump 1 step up (to the 2nd row, 1st column). This way, you can continue as shown in the figure above.

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Now we need to convert it into a set of instructions so that anyone can follow.

Instructions/Computer Program. One possible set of instructions to solve the puzzle is shown below. The starting position is the top-left square facing downwards.

1. Move 2 steps
2. Take a 180° turn
3. Move 1 step
4. Take 90° right turn
5. Take 4 steps
6. Take 90° right turn
7. Take 2 steps

Possible errors in this set of instructions include failing to mention the initial direction, students forgetting which direction they are facing, or turning in the wrong direction. When students do this activity, while they are creating their solutions, they also need to test it out and figure out the possible errors.

Explorations

The teacher needs to keep three aspects of this activity in mind.

1. Solve the puzzle: Students first have to figure out the path from the start to the finish. Once they solve it, they know which boxes are part of the solution.

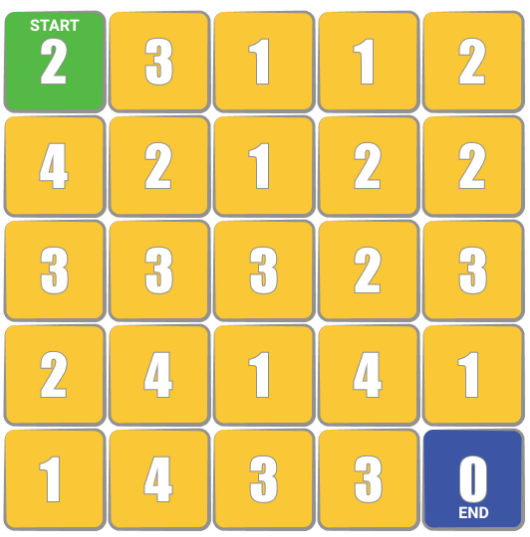
2. “Programming”: The students must now convert the path/solution into a set of instructions so that anyone can be easily guided through the maze without having to solve it themselves. This is exactly what a computer program is, in simple terms. Teachers must introduce this to their students.

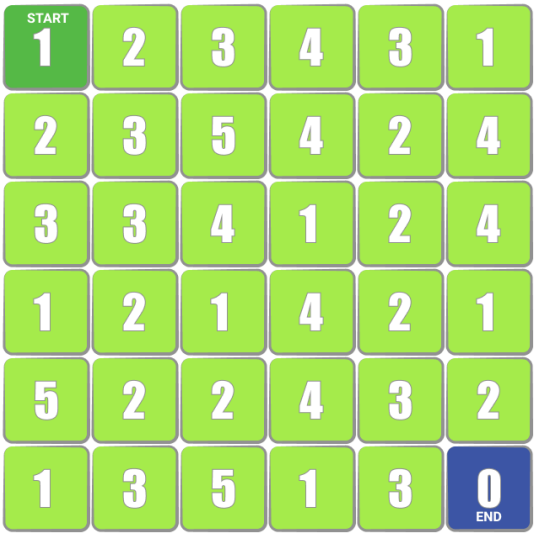
3. “Testing and correction”: As students start implementing the set of instructions by asking some of their friends to follow them, the following may occur. The initial set of instructions may not be complete, may have some inconsistencies, some errors, etc. They will have to modify their set of instructions. This is an essential part of programming that every program must be tested and corrected if required.

The teacher will give versions of number-maze puzzles for students to first solve. Once students solve it, they need to write the instructions for reaching the destination from the starting position. Let one of the group members follow the instructions as it is. If the student does not reach the destination, the group will find the mistake in the instruction and fix it. **Let the students find the solutions for the following versions of the grid puzzles. Write the instruction/computer program and test it.**

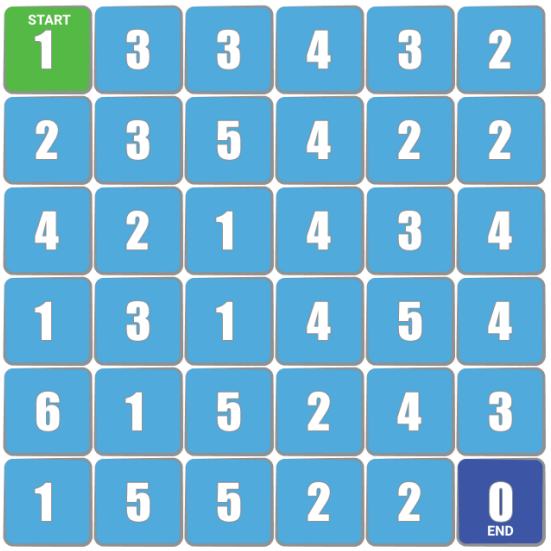
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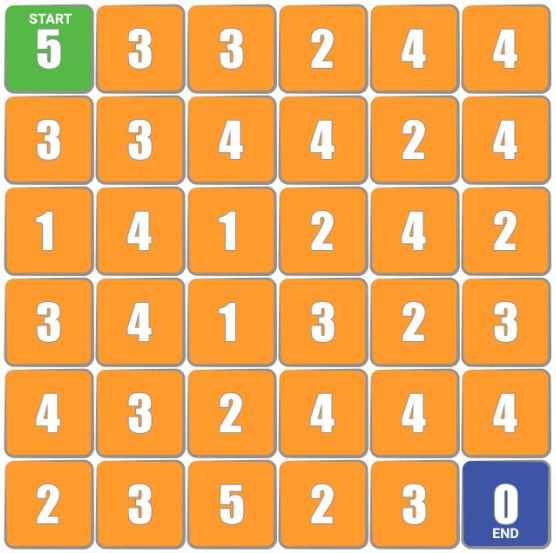
Level 1





Level 2





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